

Sirtuins in cognitive ageing and Alzheimer's disease.

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PURPOSE OF REVIEW: Sirtuins are a family of enzymes highly conserved in evolution and involved in mechanisms known to promote healthy ageing and longevity. This review aims to discuss recent advances in understanding the role of sirtuins, in particular mammalian SIRT1, in promoting longevity and its potential molecular basis for neuroprotection against cognitive ageing and Alzheimer's disease pathology. RECENT FINDINGS: Accumulative increase in oxidative stress during ageing has been shown to decrease SIRT1 activity in catabolic tissue, possibly by direct inactivation by reactive oxygen. SIRT1 overexpression prevents oxidative stress-induced apoptosis and increases resistance to oxidative stress through regulation of the FOXO family of forkhead transcription factors. In addition, resveratrol strongly stimulates SIRT1 deacetylase activity in a dose-dependent manner by increasing its binding affinity to both the acetylated substrate and NAD(+). Recently, SIRT1 has been shown to affect amyloid production through its influence over the ADAM10 gene. Upregulation of SIRT1 can also induce the Notch pathway and inhibit mTOR signalling. SUMMARY: Recent studies have revealed some of the mechanisms and pathways that are associated with the neuroprotective effects of SIRT1.